The impacts of hospital admission in very late-onset schizophrenia-like psychosis: A case report.

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Very late-onset schizophrenia-like psychosis (VLOSLP) is a psychotic disorder with an age of onset ≥60 years, and social isolation is a risk factor. Reports on the impact of interventions for isolation and loneliness on psychiatric symptoms in VLOSLP are limited. An 87-year-old woman, widowed and living alone, developed psychosis, including paranoia, erotomania, and visual hallucinations, at 84 years old during a period when her interactions with others were limited by the COVID-19 pandemic and osteoarthritis. She was eventually brought to our hospital with a local dementia outreach team. She was admitted and diagnosed with VLOSLP with mild cognitive decline through imaging and neuropsychological tests confirming the absence of dementia. Immediately after admission, her psychotic symptoms became inactive. She was transferred to another psychiatric hospital to prepare for her move to a long-term care facility because her psychosis was alleviated. During admission, she enjoyed the company of others and occupational therapy, and her score on the UCLA Loneliness Scale Version 3 improved from 44 at admission to 35 at discharge. The admission itself improved the patient's psychosis, which seemed to be related to the alleviation of isolation and loneliness.

Schizophrenia and delusional disorder in older age: community prevalence, incidence, comorbidity, and outcome.

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The opportunity to assess prevalence, incidence, and outcome of schizophrenia and delusional disorder was provided by an age- and sex-stratified random sample of 5,222 persons age 65 years and over. This sample was chosen from general practitioner lists, and interviewed by psychiatric nurses trained to use the Geriatric Mental State (GMS)-AGECAT computerized diagnostic system. GMS-AGECAT ensured the reliability of the selection of cases between the two waves of the study. A subsample was interviewed by a research psychiatrist. The sample was followed up 2 years later using the same method by interviewers blind to the initial findings. The protocols of all nominated cases and subcases of schizophrenia/paranoid disorder diagnosed by AGECAT were reviewed by a clinician and DSM-III-R diagnoses were made. Refusal rate was 13 percent for initial interviews (wave 1) and 15 percent for reinterview 2 years later (wave 2). The prevalence of DSM-III-R schizophrenia was 0.12 percent (95% confidence interval [CI] 0.04-0.25) and delusional disorder 0.04 percent (95% CI 0.00-0.14). The minimum incidence of schizophrenia for new cases was 3.0 (95% CI 0.00 to 110.70); for new and relapsed cases, 45.0 (95% CI 3.54-186.20); and for delusional disorder, 15.6 (95% CI 0.02-135.10) per 100,000 per year. Two of the five cases with schizophrenia were known to have been first diagnosed before age 65. After 2 years, none of the cases of schizophrenia had recovered fully, but none was deluded at followup. Two had developed dementia. The outcome was bad because they remained cases of some type of psychiatric illness but good because of the improvement in their schizophrenia/delusion disorder symptoms.

Socio-Cultural Factors Delaying Treatment in a Patient with Late-Onset Schizophrenia.

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Up to 23.5% of patients with schizophrenia have onset of illness after the age of 40. We report a case of a 57-year-old lady who had been sitting continuously on the toilet for 2.5 years because of persecutory delusions and somatic passivity symptoms. She was diagnosed with late-onset schizophrenia and her symptoms improved with risperidone. In this case report, we describe the phenomenology of her psychotic symptoms and explore the socio-cultural factors behind the long duration of untreated psychosis (DUP). We conclude that more can be done to improve mental health awareness and reduce the social stigma associated with mental illness.

Comparison of sociodemographic, clinical, and alexithymia characteristics of schizophrenia patients with and without criminal records.

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The primary objective of our study is to delineate differences between individuals diagnosed with schizophrenia who have a criminal record and those diagnosed with schizophrenia without a criminal record in terms of sociodemographic and clinical characteristics, levels of intelligence and insight, alexithymia, psychological symptoms, aggression, and impulsivity violence. In doing so, we aim to determine whether these findings serve as predictive indicators in the commission and prediction of criminally relevant actions in individuals diagnosed with schizophrenia. This study was conducted with patients aged 18-65 who were diagnosed with 'schizophrenia' according to DSM-5 diagnostic criteria and received outpatient follow-up and treatment. Our study consists of a total of 100 individuals diagnosed with schizophrenia, with 50 having a criminal record and 50 without. The results of the study demonstrated statistically significant differences between the forensic case and control groups in terms of gender, marital status, and educational status. Additionally, it was determined that there was a significant difference in the difficulty describing feelings between the forensic case and control groups. Statistically significant differences were obtained between the two groups in terms of BPAQ total score, the physical aggression subscale, and the anger subscale scores. There was also a statistically significant difference in terms of the BIS-11-SF total score, attention impulsivity, motor impulsivity, and non-planning subscales. Regression analysis indicated that gender, marital status, educational status, age of illness onset, difficulty in verbalizing emotions, overall aggression level, physical aggression, anger, overall impulsivity level, attention impulsivity, motor impulsivity, and inability to plan were associated with forensic behaviors in patients with schizophrenia. As a result, there is a need for studies that encompass larger and more diverse sample groups and patients from different regions. Additionally, these studies should incorporate scales and methods that comprehensively analyze both positive and negative symptoms.

Enhancer Arrays Regulating Developmental Genes: Sox2 Enhancers as a Paradigm.

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Enhancers are the primary regulatory DNA sequences in eukaryotes and are mostly located in the non-coding sequences of genes, namely, intergenic regions and introns. The essential characteristic of an enhancer is the ability to activate proximal genes, e.g., a reporter gene in a reporter assay, regardless of orientation, relative position, and distance from the gene. These characteristics are ascribed to the interaction (spatial proximity) of the enhancer sequence and the gene promoter via DNA looping, discussed in the latter part of this chapter.Developmentally regulated genes are associated with multiple enhancers carrying distinct cell and developmental stage specificities, which form arrays on the genome. We discuss the array of enhancers regulating the Sox2 gene as a paradigm. Sox2 enhancers are the best studied enhancers of a single gene in developmental regulation. In addition, the Sox2 gene is located in a genomic region with a very sparse gene distribution (no other protein-coding genes in ~1.6 Mb in the mouse genome), termed a "gene desert," which means that most identified enhancers in the region are associated with Sox2 regulation. Furthermore, the importance of the Sox2 gene in stem cell regulation and neural development justifies focusing on Sox2-associated enhancers.

Enhancer selectivity across cell types delineates three functionally distinct enhancer-promoter regulation patterns.

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Multiple enhancers co-regulating the same gene is prevalent and plays a crucial role during development and disease. However, how multiple enhancers coordinate the same gene expression across various cell types remains largely unexplored at genome scale. We develop a computational approach that enables the quantitative assessment of enhancer specificity and selectivity across diverse cell types, leveraging enhancer-promoter (E-P) interactions data. We observe two well-known gene regulation patterns controlled by enhancer clusters, which regulate the same gene either in a limited number of cell types (Specific pattern, Spe) or in the majority of cell types (Conserved pattern, Con), both of which are enriched for super-enhancers (SEs). We identify a previously overlooked pattern (Variable pattern, Var) that multiple enhancers link to the same gene, but rarely coexist in the same cell type. These three patterns control the genes associating with distinct biological function and exhibit unique epigenetic features. Specifically, we discover a subset of Var patterns contains Shared enhancers with stable enhancer-promoter interactions in the majority of cell types, which might contribute to maintaining gene expression by recruiting abundant CTCF. Together, our findings reveal three distinct E-P regulation patterns across different cell types, providing insights into deciphering the complexity of gene transcriptional regulation.

Gene activation by metazoan enhancers: Diverse mechanisms stimulate distinct steps of transcription.

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Enhancers can stimulate transcription by a number of different mechanisms which control different stages of the transcription cycle of their target genes, from recruitment of the transcription machinery to elongation by RNA polymerase. These mechanisms may not be mutually exclusive, as a single enhancer may act through different pathways by binding multiple transcription factors. Multiple enhancers may also work together to regulate transcription of a shared target gene. Most of the evidence supporting different enhancer mechanisms comes from the study of single genes, but new high-throughput experimental frameworks offer the opportunity to integrate and generalize disparate mechanisms identified at single genes. This effort is especially important if we are to fully understand how sequence variation within enhancers contributes to human disease.

Dichotomy in redundant enhancers points to presence of initiators of gene regulation.

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The regulatory landscape of a gene locus often consists of several functionally redundant enhancers establishing phenotypic robustness and evolutionary stability of its regulatory program. However, it is unclear what mechanisms are employed by redundant enhancers to cooperatively orchestrate gene expression. By comparing redundant enhancers to single enhancers (enhancers present in a single copy in a gene locus), we observed that the DNA sequence encryption differs between these two classes of enhancers, suggesting a difference in their regulatory mechanisms. Initiator enhancers, which are a subset of redundant enhancers and show similar sequence encryption to single enhancers, differ from the rest of redundant enhancers in their sequence encryption, evolutionary conservation and proximity to target genes. Genes hosting initiator enhancers in their loci feature elevated levels of expression. Initiator enhancers show a high level of 3D chromatin contacts with both transcription start sites and regular enhancers, suggesting their roles as primary activators and intermediate catalysts of gene expression, through which the regulatory signals of redundant enhancers are propagated to the target genes. In addition, GWAS and eQTLs variants are significantly enriched in initiator enhancers compared to redundant enhancers, suggesting a key functional role these sequences play in gene regulation. The specific characteristics and widespread abundance of initiator enhancers advocate for a possible universal hierarchical mechanism of tissue-specific gene regulation involving multiple redundant enhancers acting through initiator enhancers.

Enhancers and super-enhancers have an equivalent regulatory role in embryonic stem cells through regulation of single or multiple genes.

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Transcriptional enhancers are critical for maintaining cell-type-specific gene expression and driving cell fate changes during development. Highly transcribed genes are often associated with a cluster of individual enhancers such as those found in locus control regions. Recently, these have been termed stretch enhancers or super-enhancers, which have been predicted to regulate critical cell identity genes. We employed a CRISPR/Cas9-mediated deletion approach to study the function of several enhancer clusters (ECs) and isolated enhancers in mouse embryonic stem (ES) cells. Our results reveal that the effect of deleting ECs, also classified as ES cell super-enhancers, is highly variable, resulting in target gene expression reductions ranging from 12% to as much as 92%. Partial deletions of these ECs which removed only one enhancer or a subcluster of enhancers revealed partially redundant control of the regulated gene by multiple enhancers within the larger cluster. Many highly transcribed genes in ES cells are not associated with a super-enhancer; furthermore, super-enhancer predictions ignore 81% of the potentially active regulatory elements predicted by cobinding of five or more pluripotency-associated transcription factors. Deletion of these additional enhancer regions revealed their robust regulatory role in gene transcription. In addition, select super-enhancers and enhancers were identified that regulated clusters of paralogous genes. We conclude that, whereas robust transcriptional output can be achieved by an isolated enhancer, clusters of enhancers acting on a common target gene act in a partially redundant manner to fine tune transcriptional output of their target genes.

Analysis of single-cell CRISPR perturbations indicates that enhancers predominantly act multiplicatively.

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A single gene may have multiple enhancers, but how they work in concert to regulate transcription is poorly understood. To analyze enhancer interactions throughout the genome, we developed a generalized linear modeling framework, GLiMMIRS, for interrogating enhancer effects from single-cell CRISPR experiments. We applied GLiMMIRS to a published dataset and tested for interactions between 46,166 enhancer pairs and corresponding genes, including 264 "high-confidence" enhancer pairs. We found that enhancer effects combine multiplicatively but with limited evidence for further interactions. Only 31 enhancer pairs exhibited significant interactions (false discovery rate <0.1), none of which came from the high-confidence set, and 20 were driven by outlier expression values. Additional analyses of a second CRISPR dataset and in silico enhancer perturbations with Enformer both support a multiplicative model of enhancer effects without interactions. Altogether, our results indicate that enhancer interactions are uncommon or have small effects that are difficult to detect.